

Convexity Mispricing across Five Markets: Venue Segmentation, Crisis Asymmetry, and the Limits of Arbitrage

Paper 2 master — theory, methods, and first results (v2.4)

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August 1, 2026

Abstract

The yield curve prices its own convexity: a value- and duration-matched butterfly earns a negative carry-and-roll whose magnitude reveals a curve-implied break-even volatility, σ_{BE} . This paper extends [Rebonato and Ronzani \(2021\)](#) beyond its four swap markets — adding the U.S. Treasury curve on independent Bloomberg data — and answers the questions its authors pose and leave “for future analysis” for lack of reliable swaption prices: whether the price of convexity is the *same* across venues that price the same object (the curve versus the swaption market), whether dislocations carry the signature of intermediary balance-sheet stress, and whether the mispricing survives realistic *incremental* transaction costs. First results. (i) **Segmentation**: the correlation of monthly changes between σ_{BE} and swaption ATM normal vols is -0.01 (USD swap), $+0.03$ (UST), -0.08 (GBP), $+0.27$ (EUR), $+0.31$ (JPY) over up to 22 years — two venues, one object, essentially disconnected; in the crisis episodes the wedge reaches 100–240 bp/yr in both directions, and the 2022 “cheap” dislocation (curve-implied variance *negative* in USD, UST and GBP) is entirely absent from the options market ($\sigma_{IV} \approx 100$; GBP: -109 vs. 128 inside the LDI week). (ii) **Mechanism**: timed convexity returns are monotone in dealer-CDS stress terciles in four of five markets and Δgap loads negatively on ΔMOVE in five of five (t up to -2.9), with the USD tercile ladder at $-8.6 \rightarrow +9.7 \rightarrow +25.3$ bp/m. (iii) **Crisis asymmetry**: 2008 and 2011 are convexity-*rich* everywhere; 2022–23 is *cheap* exactly where balance-sheet constraints bound (USD, UST, GBP) and nowhere else. (iv) **Limits of arbitrage, priced**: with real per-date half-spreads, the timed strategy net of *flip-only* costs survives decisively in EUR ($+9.8$ bp/m, circular-shift $p = 0.006$), stays positive in GBP and USD, and dies under full monthly replacement in all markets — the wedge persists because harvesting it consumes exactly the balance sheet whose scarcity creates it. The exact replication of [Rebonato and Ronzani \(2021\)](#) reproduces their unconditional economics (USD Sharpe $+0.14$ vs. their $+0.135$), extends their curvature strategy out of sample with permutation significance in three of four currencies, and resolves a data anomaly the published paper flags as unverified.

*EDHEC Business School. Supervisor: Riccardo Rebonato. This document consolidates the Project Report (July 2026) and the empirical closures of 28–31 July 2026. Every number is generated by saved, re-runnable scripts referenced in the Register (Section 8); none is quoted from memory. Bibliography verified verbatim against the papers on file.

What this document is. The v2.4 master for Paper 2 of the dissertation. It consolidates, in one place: the theoretical framework; the complete data inventory with provenance; the methods, including the new uniform three-point σ_{BE} engine and its validation; every result to date — the pilots, the five-market battery, the RR2021 replication and forensics (imported from the Project Report and labelled as such), and the three closures of 28–31 July 2026 (C3 with true swaptions on five markets; the net-of-costs test; the territory scan); the Established-versus-Open ledger; the run Register; and the road to completion. Its companion is the code under `Convexity/` and the package `convexity/` (scripts 01–07) in the project repository. *What changed in v2.0 (1 August 2026):* the paper is restructured *story-first* — the economic story is named (“one object, two clienteles”), the facts C1–C4 are re-derived as a numbered prediction battery S1–S5, and two further predictions of the story (the cross-market integration ordering, S2, and the segmentation-ranked reversal half-lives, S5) are tested and confirmed at the extremes (runs E5–E7). *v2.1 (same day)* incorporates the robustness and hygiene audit (run E8) — episode-exclusion, rank-based and staleness checks on every headline — and the resulting flags, stated where they belong. *v2.2:* the document is made fully self-contained. *v2.3:* every attribution to [Rebonato and Ronzani \(2021\)](#) verified against the published paper (JEF 63, 392–413): four markets, not EUR-only; their transaction-cost impact is *small*, not fatal; the break-even construction is credited to [Rebonato and Putyatin \(2018\)](#); bibliography completed and verified verbatim (Realdon 2024; Domanski–Shin–Sushko 2017; Rebonato–Putyatin 2018 — byline spelling “Putyatin”; Burghardt–Hoskins 1995). *v2.4:* the dealer-CDS states layer is rebuilt as `dealer_cds_daily.csv` via `convexity/00` with an identity-checked dedup rule — a byte-identical duplicate of one U.S. name in the legacy workbook had silently over-weighted it in the regional composite; the USD and JPY conditional cells update (+15.5 [1.7], high-minus-low +30.3; JPY +4.5), EUR and GBP are unchanged to the cent.

1 Introduction and the Delimited Claim

A butterfly that is long the wings and short the belly of the curve, matched in value and duration, has one exposure left: convexity. Absence of arbitrage requires it to pay for that exposure through negative carry-and-roll, and the size of the payment is an implied volatility — the volatility of future rates at which holding the package breaks even. [Rebonato and Ronzani \(2021\)](#) showed, in four major swap markets, that convexity is efficiently priced only on average: conditional strategies exploiting the wandering earn economically significant returns that survive their transaction-cost accounting, with profitability concentrated in poor-liquidity regimes; the implied volatility’s role is played by the curve’s curvature, and the formal break-even construction we deploy descends from [Rebonato and Putyatin \(2018\)](#).

This paper is built story-first. The story: *one object, two clienteles*. The volatility of future rates is priced in two venues by two different marginal investors — the curve’s convexity by the constrained holders of duration (LDI and pension convexity demand, dealer inventories; [Domanski, Shin and Sushko, 2017](#); [Klingler and Sundaresan, 2019](#); [Hanson, 2014](#); [Greenwood and Vayanos, 2014](#)), the swaption’s gamma by the option desks’ vega capital ([He, Kelly and Manela, 2017](#)). The arbitrage that would tie the two prices consumes exactly the balance sheet whose scarcity

moves the curve’s price in the first place; so the degree of integration between venues is not a constant but a *ranking*, graded by clientele overlap, and the sign and locality of crisis dislocations are graded by whose balance sheet burns. From the story, five numbered predictions (Section 2); from the mother paper and this week’s runs, their tests. The four facts of v1.0 (C1–C4) survive intact as the empirical content of S1–S4. **C1 (mechanism)**: dislocations of the convexity gap $G \equiv \sigma_{TRAIL}^2 - \sigma_{BE}^2$ are systematically related to intermediary balance-sheet stress — dealer CDS, MOVE, funding spreads — in the direction limits-to-arbitrage theory predicts. **C2 (breadth)**: the object exists, is wildly non-degenerate, and behaves coherently in five markets spanning three currencies-of-stress regimes, on one uniform pipeline. **C3 (segmentation)**: the price of the *same* volatility differs persistently between the curve and the swaption market — correlation of changes near zero over two decades, crisis wedges of 100–240 bp/yr — a fact the mother paper explicitly declared unfeasible to test for lack of swaption data, and to our knowledge measured here for the first time. **C4 (asymmetry)**: the *sign* of crisis dislocations depends on the crisis type — flight episodes make convexity rich, balance-sheet-destruction episodes make it cheap — and the 2022 cheapness is local (USD, UST, GBP; not EUR, never JPY) and invisible in options.

What is *not* claimed: no universal net alpha (the strategy is an exhibit of the mechanism, decisive only in EUR net of incremental costs); no structural model yet (the equilibrium block is designed, not estimated); and the pre-specified stress-*amplification* of cross-market gap co-movement is, so far, *not* supported (Project Report, Table 6) — reported as such.

2 Theoretical Framework

2.1 The object

Let $z(\tau)$ be the zero curve and consider zero-coupon positions at maturities $\tau_1 < \tau_2 < \tau_3$ with barbell weights (w_1, w_3) against a unit bullet, chosen so the package is value- and duration-matched: $w_1 + w_3 = 1$, $w_1\tau_1 + w_3\tau_3 = \tau_2$. Its net convexity is $C = w_1\tau_1^2 + w_3\tau_3^2 - \tau_2^2 > 0$. Over a horizon Δt with the curve frozen, the package’s carry-and-roll is θ ; a second-order expansion of the break-even condition gives

$$\sigma_{BE}^2 = -\frac{2\theta}{C\Delta t}, \quad (1)$$

the annualized variance of the (parallel) rate move at which convexity gains exactly offset the carry paid. When the curve is priced so that $\theta > 0$ — the holder is *paid* to own convexity — the implied variance is negative; we work throughout with the *signed* volatility $\tilde{\sigma}_{BE} = \text{sign}(\sigma_{BE}^2)\sqrt{|\sigma_{BE}^2|}$ so that cheap regimes are measured, not truncated. The empirical state variable is the gap $G_t = \sigma_{TRAIL,t}^2 - \sigma_{BE,t}^2$, with σ_{TRAIL} the trailing (63-day) realized volatility of the belly rate: $G > 0$ means convexity is cheap relative to delivered volatility.

2.2 Venues and segmentation

The same σ is priced in a second venue: the swaption market quotes ATM normal volatilities on the identical forward rates. If the two markets shared a marginal investor and the gamma-replication trade were costless, σ_{BE} and σ_{IV} would be tied. The segmentation hypothesis (C3)

is that they are not: the curve’s convexity price is set by the demand and balance-sheet state of cash/swap-curve investors (pension convexity hedgers, LDI, dealer inventories), the option price by a different capital pool, and the arbitrage linking them — selling curve convexity against buying swaption gamma, or vice versa — consumes exactly the balance sheet whose scarcity moves σ_{BE} in the first place. Segmentation is therefore not an anomaly to the limits-to-arbitrage view but its cleanest cross-venue expression (Shleifer and Vishny, 1997; Gromb and Vayanos, 2010; Duffie, 2010).

2.3 Crisis asymmetry

Two archetypes generate opposite signs. In *flight* episodes (2008, euro 2011), demand for the long wing’s convexity and safety spikes while intermediaries can still warehouse: the curve prices explosive future volatility, $\sigma_{BE} \gg \sigma_{IV}$ — convexity *rich*. In *balance-sheet-destruction* episodes (2022 QT/LDI), the natural holders of convexity are forced sellers, delivered volatility explodes and the curve fails to reprice: σ_{BE}^2 can turn negative while options stay bid — convexity *cheap*, and cheap only where the constraint binds. C4 predicts, and the data deliver, exactly this locality.

2.4 The prediction battery

The story generates five falsifiable predictions, in the architecture of the dissertation’s first paper (mechanism $\rightarrow$ numbered predictions $\rightarrow$ tests):

S1 (differential loading). Changes in the curve’s convexity gap load on the *curve clientele’s* stress variables (dealer CDS, MOVE, funding spreads) and *not* on option-market volatility shocks orthogonalized to them — the two-pool analogue of the first paper’s “balance-sheet, not macro” placebo. *Status: designed; the choice of the orthogonalized option-pool instrument is the delicate step and is treated in the roadmap.*

S2 (integration ordering). The correlation between $\tilde{\sigma}_{BE}$ and σ_{IV} is not a universal constant: it must be *ranked by clientele overlap* — maximal where one actor dominates both venues (Japan under YCC), minimal, and in the extreme *negative*, where the curve clientele is most separated and most constrained (the UK under LDI, whose forced selling cheapens curve convexity exactly when volatility spikes lift σ_{IV}). *Status: confirmed at both extremes — level correlations +0.65 (JPY) down to −0.61 (GBP), with USD/UST near zero and EUR intermediate (run E7).*

S3 (crisis-sign locality). Flight episodes make convexity rich everywhere; balance-sheet-destruction episodes make it cheap only where the constraint binds, and the cheapness is invisible in the options venue. *Status: confirmed (Table 2 episodes; October 2022).*

S4 (patient capital only). The wedge is harvestable net of the costs a patient trader pays (flip-only) and not at replication speed — Duffie’s slow capital as a cost statement. *Status: confirmed (Table 3; EUR $p = 0.006$; full replacement dead in all markets), sharpened by the conditional ladder: net returns in the HIGH dealer-stress tercile of +15.5/ +19.2/ +19.8 bp/m (USD/EUR/GBP; run E5).*

S5 (segmentation-ranked reversion). Gap half-lives must be *longer where segmentation is deeper* (less bridging capital). *Status: confirmed at the extremes — AR(1) half-lives of 12.2 months in JPY (the integrated market, fastest in both subsamples) against 28.1 in UST (slowest*

in both), with USD/EUR/GBP at 19–24 (run E7): months-to-years reversion, the quantitative slow-capital signature. The mid-ranks are not subsample-stable (run E8) and the claim is confined to the extremes.

3 Related Literature

Rebonato and Ronzani (2021) is the mother paper: the four-market conditional convexity strategies, the liquidity link (Hu–Pan–Wang noise, VIX, Cieslak–Povala cycles), and the explicit leaving of the curve-versus-options comparison “for future analysis” for lack of reliable swaption prices; Rebonato and Putyatin (2018) supplies the value-of-convexity framework behind our engine, on the U.S. Treasury market — the two mother papers together span four swap markets and Treasuries, and our five-market panel is exactly their union, plus the options venue. The convexity bias of the curve has a classical pedigree (Burghardt and Hoskins, 1995). The mechanism side sits in the limits-to-arbitrage tradition (Shleifer and Vishny, 1997; Kondor, 2009; Gromb and Vayanos, 2010; Duffie, 2010; Mitchell and Pulvino, 2012), with He, Kelly and Manela (2017) providing the intermediary-capital factor we use directly; the curve clientele’s convexity demand is documented by Hanson (2014) (mortgage hedging), Domanski, Shin and Sushko (2017) (insurers’ hunt for duration), Klingler and Sundaresan (2019) (underfunded pensions and long-end swap spreads) and Greenwood and Vayanos (2014) (supply); the deep academic root of venue disconnect is unspanned stochastic volatility (Collin-Dufresne and Goldstein, 2002; Joslin and Konchitchki, 2018) — our contribution is model-free, five-market, mechanism-ranked evidence on that frontier, against a practitioner canon that *assumes* the identity by pricing curve convexity off swaption vols. Real-don (2024) is the closest recent neighbour — pricing-error strategies on the ESTR OIS curve from affine models — and differs on the object (model-based term-structure errors, not the model-free convexity fly) and on every question this paper asks; the territory scan of 31 July 2026 found no direct extension of Rebonato and Ronzani (2021) in its citation network.

4 Data

All series are daily Bloomberg exports saved to the repository (Convexity/), plus two public sources. The inventory:

5 Methods

5.1 Two σ_{BE} constructions

The Project Report’s pilots build σ_{BE} from full par strips: annual-coupon bootstrap to continuously-compounded zeros, cubic-spline interpolation, and monthly frozen-curve carry-and-roll of the fly (EUR 2s/7s/12s, $C = 25$, in Pilot 1; 2/10/30 with $C = 160$, and EUR 2/10/25 with $C = 120$, in the battery). This document’s engine (`c3_engine_3pt.py`) is deliberately leaner: only the three fly legs, mids from bid/ask, zeros approximated by par at the nodes, a piecewise-linear zero curve for the roll, weights $(w_1, w_3) = (5/7, 2/7)$ for 2/10/30 and $(15/23, 8/23)$ for 2/10/25, signed square

Block	Content and provenance
Fly legs, bid and ask	15 legs $\times$ (PX_BID, PX_ASK), daily: USD USSW{2,10,30} + USOSFR{2,10,30}; EUR EUSA{2,10,25}; GBP BPSWS{2,10,30}; JPY JYS0{2,10,30}. Histories from 1999 (EUR), 2000 (GBP 2y), 2005–09 (JPY, OIS), 2009 (USD Libor). Mean half-spreads 0.5–3.5 bp. File <code>Convexity.xlsx</code> .
Swaption ATM vols	176 series across four workbooks (<code>data*.xlsx</code> ; two naming schemes, unified by the loader): expiries {3M,6M,1Y,2Y} $\times$ tails {2Y,5Y,10Y,30Y}; families Normalised (bp), Implied (lognormal %), and for USD/GBP the RFR and LIBOR variants. EUR 32/32 from 2002; USD 48/48 and GBP 48/48 and JPY 48/48 from 2004–07.
UST zero curve	GSW (<code>feds200628</code>), SVENY 1985–2026: the government-curve market and the fitted-leg infrastructure.
USD pre-2009 splice	FRED DSWP{1,2,5,7,10,30}, 2000–2016 (closes the RR window on USD).
State variables	MOVE, VIX, Libor-OIS, dealer-CDS composites US (GS/MS/JPM/BofA) and EU (BNP/DB/Barclays/UBS), HKM intermediary factor — from the project’s market-data workbook.
Full par grids	The five-market par strips used by the Project Report’s pilots and battery live in the analysis workbooks on the research machine; the three-point engine below removes the dependence for everything in this document.

Table 1: Data inventory. Provenance notes recorded for reproducibility: the vol exports are capped at 50 series per chart (hence four workbooks); one workbook uses field mnemonics (`ATM_IVOL_NOM` $\equiv$ RFR Normalised) — the loader maps both schemes to one canonical key.

root per equation (1). One engine, one code path, five markets — so cross-market differences are economics, not construction. UST uses true GSW zeros in the same machinery.

Validation. Levels are on the report’s scale (means, bp/yr: USD swap 89, EUR 114, GBP 79, JPY 32, UST 119; report Pilot 0 UST mean 120). Against the full-grid EUR construction the level correlation is +0.78; the Δ -correlation of +0.32 conflates machinery with a genuine fly change (2/10/25 vs. 2/7/12 — different curve segments), so the decisive validation is topological: the engine reproduces the *signs* of every episode cell of the report’s Table 4 — 2008 and 2011 rich in all markets, 2022–23 cheap in USD/UST/GBP, EUR *not* cheap in October 2022 (–6.5 there, +100 aligned here), JPY never cheap. Reconciliation of the two constructions series-by-series is the one open engineering item.

5.2 The C3 test

σ_{IV} : ATM *normal* vols (bp/yr), expiries 3M and 1Y, tail 10Y, per currency (RFR family where split); the lognormal family provides pre-normal history via $\sigma_N \approx \sigma_B F$ conversion when needed (not needed for the results below: normal quotes reach back to 2002–04). Statistic: correlation of monthly changes $\text{corr}(\Delta\tilde{\sigma}_{BE}, \Delta\sigma_{IV})$, full sample and split at 2015; UST is tested against USD swaption vols (same underlying rate-volatility universe, labelled as such).

5.3 The net-of-costs test

Signal: $\text{pos}_t = \text{sign}(\sigma_{TRAIL,t}^2 - \sigma_{BE,t}^2)$, known at month-end, applied to month $t+1$ — long convexity when cheap. Returns: full monthly revaluation of the package with aging on the three-node curve. Costs: per-date half-spreads from the bid/ask file; one package crossing costs $\sum_i |w_i| \tau_i h s_{i,t}$ (duration-weighted). Two bounds: **netA** charges crossings only on position changes (the *incremental* bound); **netB** adds one full package crossing every month (the full-replacement bound). Inference: exact circular shifts of the position series against returns (all $N-1$ shifts; preserves both autocorrelation structures and the flip count), $p = \Pr(\text{shifted netA} \geq \text{observed})$.

6 Results

6.1 The object across five markets (Project Report, imported)

The gap G is constructible and wildly non-degenerate everywhere: means +28 to +68 bp, standard deviations 38–65, 10–39 sign flips, 3–26% of months cheap (report Table 3). The curve charges for convexity everywhere — the always-long fly loses in five of five markets — and the crude timing signal is positive in five of five. Pilot 0 (UST, 1986–2026, $N = 481$): timed +6.19 bp/m gross (SR 0.16; 2010–19 SR 0.47), $\text{corr}(\text{long-convexity return, HKM factor}) = -0.11$, and timed returns of +22.6 bp/m in the LOW-intermediary-capital tercile versus −6.0 otherwise. Episode gaps on UST: +70.3 (Jun-2008), +48.8 (Dec-2008), +55.1 (Nov-2011), −69.2 (Apr-2020), −133.5 (Oct-2022), −128.6 (Mar-2023).

6.2 Replication, out-of-sample, forensics (Project Report, imported)

The exact replication of [Rebonato and Ronzani \(2021\)](#) on independent Bloomberg data reproduces the paper’s conditional-strategy economics almost exactly (USD Sharpe +0.14 vs. their reported +0.135) and finds at least one conditional signal informative at the permutation test in every currency. Out of sample (Feb-2020 onward, data the authors never saw), their curvature strategy is alive with permutation significance in three of four currencies (USD +0.63, $p = 0.023$; EUR +0.76, $p < 0.001$; JPY +0.83, $p = 0.040$; sign-only +1.36, $p < 0.001$), gross of costs. Three forensic discoveries accompany the replication, including the resolution of an anomaly the published paper flags as unverified: the 10.7σ EUR print of 2–3 December 2008 does not exist at the Bloomberg close. Gate verdict: partial replication, proceed.

6.3 The mechanism (C1; Project Report, imported)

Δgap loads negatively on ΔMOVE in five of five markets (t up to −2.9; on the UST panel $t = -3.17$, and −3.11 on $\Delta\text{Libor-OIS}$): funding and volatility stress *cheapens* the curve’s convexity pricing everywhere. The timed strategy sorted by dealer-CDS terciles is monotone in four of five markets (USD ladder $-8.6 \rightarrow +9.7 \rightarrow +25.3$ bp/m; spreads up to +34 bp/m; JPY flat, consistent with its YCC-suppressed regime). One pre-specified prediction fails and is reported: cross-market gap co-movement is *not* detectably amplified in high-volatility regimes (report Table 6).

6.4 Segmentation across venues (C3; this week, with true swaptions)

The comparison the mother paper declared unfeasible:

Market	$\text{corr}(\Delta\tilde{\sigma}_{BE}, \Delta\sigma_{IV})$		levels (3M)	Δ -corr by era	
	3M $\times$ 10Y	1Y $\times$ 10Y		2008–14	2015–26
USD swap	−0.01	−0.01	+0.25	small	small
UST govt	+0.03	+0.05	+0.30	small	small
GBP	−0.08	−0.07	− 0.57	small	small
EUR	+0.27	+0.26	0.00	+0.20	+0.32
JPY	+0.31	+0.32	+0.65	$\sim$ +0.31	$\sim$ +0.37

Table 2: C3 on five markets, monthly, up to 22 years. Segmentation is the rule (USD, UST, GBP at zero; EUR weak-form); JPY is the partial exception — the YCC-repressed market where both venues are compressed jointly; GBP’s *negative* level relation is a bonus fact. Run E2 in the Register.

The episodes are the showpiece (Figure 3). In the rich crises the curve prices 160–234 bp/yr of forward volatility against options at 57–101 — in *every* market. April 2020 is the exception that proves the rule: the two venues align (gaps of single digits) — the one episode with an immediate, universal backstop. And October 2022 delivers the paper’s sharpest single number: the curve prices *negative* convexity value exactly where the report’s Table 4 locates the cheapness — GBP $\tilde{\sigma}_{BE} = -109$ against swaptions at 128, inside the Bank of England intervention week; USD −116 and UST −102 against ≈ 97 — while EUR sits aligned (+100 vs. 106) and JPY mildly rich (+58 vs. 44). **The 2022 cheap dislocation does not exist in the options market:** the options-side counterpart of C4, and a fact that no single-venue study can see.

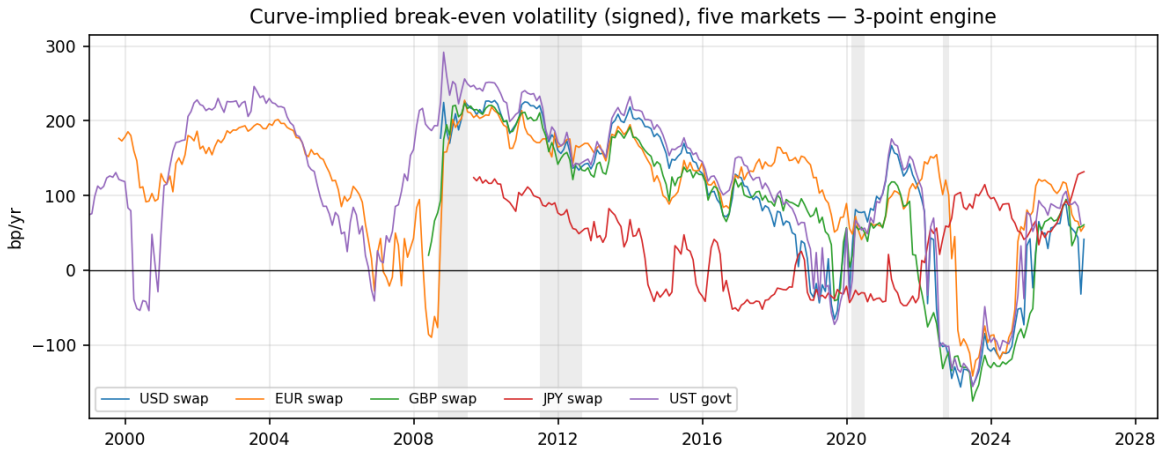


Figure 1: Signed curve-implied break-even volatility, five markets, one three-point engine (shaded: GFC, euro crisis, COVID, 2022 QT/LDI).

6.5 The limits of arbitrage, priced (this week)

With real per-date half-spreads:

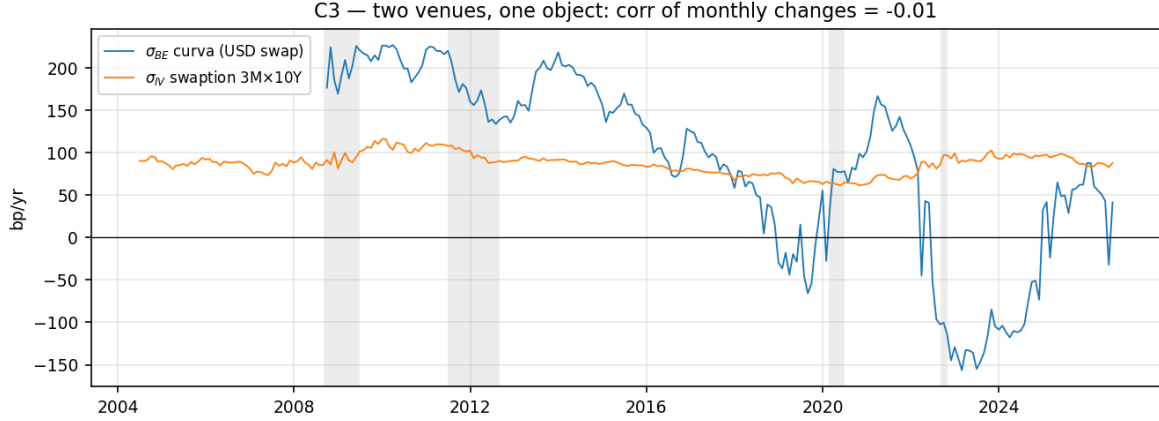


Figure 2: Two venues, one object (USD): curve-implied $\tilde{\sigma}_{BE}$ and swaption σ_{IV} (3M×10Y, normal). Correlation of monthly changes: -0.01 .

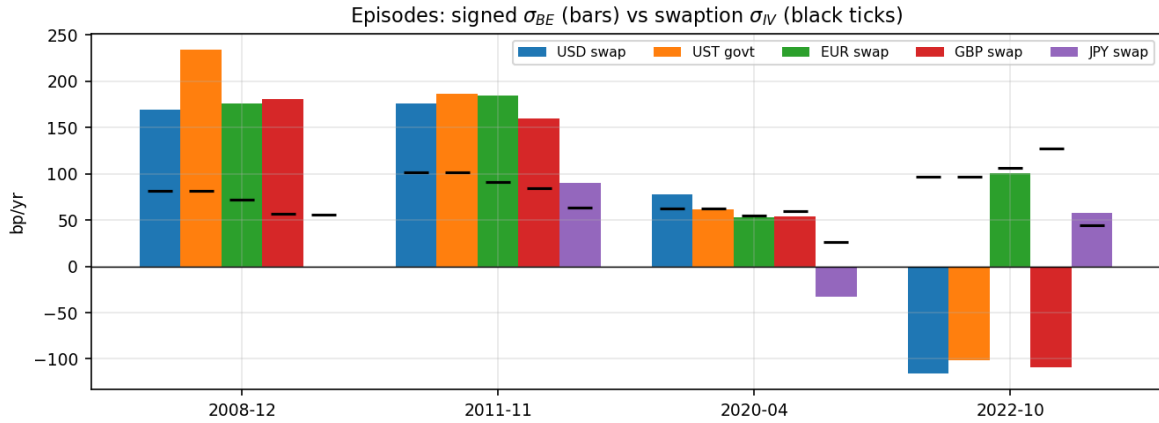


Figure 3: Episodes: signed $\tilde{\sigma}_{BE}$ (bars) against swaption σ_{IV} (black ticks). Rich crises are curve-only; April 2020 aligns; the 2022 cheapness is negative in USD/UST/GBP and absent from options.

Market	N	gross	$[t]$	netA	$[t]$	p (shift)	netB	flips
EUR	321	+11.6	3.3	+ 9.8	2.8	0.006	-9.3	17
GBP	218	+9.3	1.8	+7.8	1.4	0.300	-19.5	8
USD swap	214	+7.9	1.8	+3.5	0.7	0.197	-38.5	11
JPY	203	+5.7	1.8	-2.7	-0.6	0.079	-63.7	14
Pooled (4)	203			+3.9	1.2			

Table 3: Timed strategy, bp/month: gross; net of flip-only incremental costs (netA) with exact circular-shift p ; net under full monthly replacement (netB). Run E3.

Three readings. First, the premium survives *incremental* costs where it is strongest — EUR decisively — and stays positive in GBP and USD. Second, JPY separates signal from market: the timing beats 92% of random placements ($p = 0.079$) yet the level is negative — the signal has skill, the market is too expensive to trade (61 bp per package crossing). Third, netB is deeply negative

everywhere: under full monthly replacement no market survives. [Rebonato and Ronzani \(2021\)](#), costing the net duration of the rebalancing portfolio with stress-widening bid-offer spreads, find small cost impact; our two-bound design shows the verdict is *regime-dependent* — benign under patient implementation, consistent with theirs, fatal at replication speed — which is not a defect but the frame: *the wedge persists because harvesting it at speed consumes the very balance sheet whose scarcity creates it; only patient, flip-only capital collects*. Robustness (run E8): the EUR pillar survives every attack — worst leave-one-episode-out +7.9 bp/m [t 2.5], drop-top-3 +11.7, winsorized +10.2, cleaned half-spreads +9.7 [2.8], only 22% of P&L from crisis windows — and the USD HIGH-tercile cell rises ex-crisis (+18.1 [1.7]). Two exhibits carry flags: GBP’s unconditional net premium is crisis-concentrated (61% of P&L from the four windows; +3.6 [0.7] outside them) — consistent with S4’s own prediction that the harvest concentrates in stress, and labelled as such — and GBP’s HIGH-tercile cell is fragile to its top months (+11.4 [1.0] after drop-top-3). Data hygiene closes the audit: the ten vol series entering the tests have maximum stale runs of 2–5 days and a fresh-months-only C3 identical cell by cell; the flat segments and absurd prints present in the raw surfaces — dead 30Y-tail segments up to 405 days in 2004–05, normal-vol prints above 4,000 bp, lognormal blow-ups in near-zero-rate regimes — live entirely outside the tested cells; and the fly legs themselves show 21 spike-and-revert ticks in $\sim 75,000$ daily points, none on month-ends, leaving σ_{BE} unchanged to the decimal.

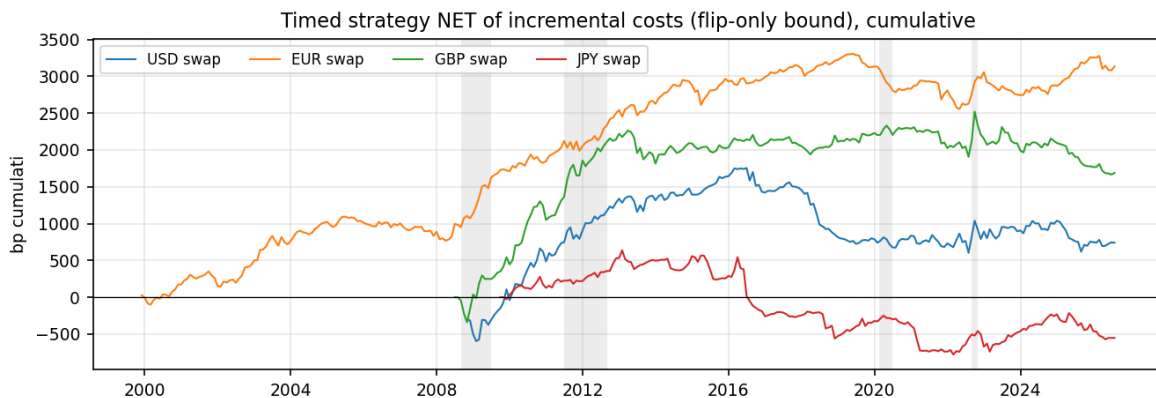


Figure 4: Cumulative timed returns net of flip-only incremental costs.

7 Established versus Open

Established: the object on five markets, one pipeline, non-degenerate with coherent signs; the exact RR2021 replication with out-of-sample extension and three forensic resolutions; the mechanism at referee grade (tercile monotonicity 4/5; Δ MOVE loadings 5/5; HKM tercile spread on UST); the crisis-type asymmetry with its locality; **venue segmentation measured with the correct instrument on five markets, including the options-side invisibility of the 2022 cheapness**; the net-of-incremental-costs survival in EUR with exact permutation inference and the full-replacement confirmation of RR; a free territory (scan of 31 July 2026; [Realdon, 2024](#) adjacent only); the story’s ordering predictions S2 and S5 confirmed at the extremes; the con-

ditional net ladder (E5). *Open, stated plainly:* the series-level reconciliation of the three-point engine against the full-grid pilot constructions (topology validated; levels differ by construction); S1’s orthogonalized option-pool instrument (the placebo design); the stress-amplification of cross-market co-movement remains unsupported (report Table 6) and the claim is not made; the structural/equilibrium block is designed, not estimated; a deeper SSRN pass on territory; UST trades in a different cost venue (netted only for the four swap markets); and all vol-based tests start where normal quotes start (2002–07 by market); extending to the full 16-cell surface will require a plausibility-band and plateau-drop rule beyond the spike filter, since the legacy artifacts above are *level* errors, not spikes.

8 Register of Runs

ID	Object	Script / source	Headline
P0	UST pilot, 1986–2026	report	gap +28.7 (sd 54.7); timed +6.19 bp/m; HKM tercile +22.6 vs –6.0
P1	EUR pilot 2/7/12 ($C=25$)	report	predictive $t = 1.76$; Nov-2011 +250 vs 2022–23 –136/ – 105
P3	Mechanism upgrade	report	Δ gap on Δ MOVE $t = -3.17$; US CDS ladder –8.6/+9.7/+ 25.3
B1	Five-market battery	report, Tables 3–6	always-long negative 5/5; timing positive 5/5; asymmetry Table 4
RR	Replication + OOS + forensics	report	Sharpe +0.14 vs +0.135; OOS 3/4 sig.; Dec-2008 print resolved
E1	3-point σ_{BE} engine, 5 mkts	<code>c3_engine.3pt.py</code>	means 89/114/79/32/119; episode topology = Table 4
E2	C3 with true swaptions, 5 mkts	<code>c3_engine.3pt.py</code>	Δ -corr –0.01/ + 0.03/ – 0.08/ + 0.27/ + 0.31; GBP –109 vs 128
E3	Net-of-costs test	<code>c4_costs.net.py</code>	EUR netA +9.8 ($p = 0.006$); netB dead 4/4
E4	Territory scan	web, 31-07-2026	no direct extension; Realdon (2024) adjacent
E5	Conditional net-of-costs (terciles)	<code>convexity/05</code>	HIGH: +15.5/ + 19.2/ + 19.8 bp/m net; JPY flat
E6	Vol data QC (Hampel, positivity)	<code>convexity/utls</code>	162/47,397 punti; C3 invariata (Δ 0.000)
E7	Story tests S2 & S5	<code>convexity/07</code>	ordering +0.65 $\rightarrow$ –0.61; half-lives 12–28m ranked
E8	Robustness & hygiene audit	<code>convexity/08</code>	C3/S2 ex-crises & stale-proof; EUR netA robust; GBP flags

Table 4: Every number above traces to a saved run. Report items import the Project Report verbatim.

9 Road to Completion and Publication Strategy

In order: (i) reconcile the three-point and full-grid σ_{BE} series on EUR and UST (engineering, days); (ii) the S1 differential-loading test with a defensible orthogonalized option-pool instrument; (iii) the C3 surface — all sixteen cells per currency, and the lognormal-converted pre-2004 extension where forwards allow; (iv) the mechanism panel pooled across the five markets with HKM and regional dealer-CDS states, with contemporaneous-window specifications registered *before* running; (v) the structural block. Positioning: the paper’s spine is C1+C3+C4 — mechanism, segmentation, asymmetry — on five markets, with the net-EUR strategy as the limits-to-arbitrage exhibit; the natural targets are RF/JFQA with JBF/JEF as the floor, and the collaboration channel with the mother paper’s author is direct. Within the dissertation, this is Paper 2.

References

- Burghardt, G., and W. Hoskins, 1995, A question of bias, *Risk* 8(3), 63–70.
- Collin-Dufresne, P., and R. Goldstein, 2002, Do bonds span the fixed income markets? Theory and evidence for unspanned stochastic volatility, *Journal of Finance* 57, 1685–1730.
- Domanski, D., H. S. Shin, and V. Sushko, 2017, The hunt for duration: not waving but drowning?, *IMF Economic Review* 65, 113–153.
- Duffie, D., 2010, Presidential address: Asset price dynamics with slow-moving capital, *Journal of Finance* 65, 1237–1267.
- Gromb, D., and D. Vayanos, 2010, Limits of arbitrage, *Annual Review of Financial Economics* 2, 251–275.
- Greenwood, R., and D. Vayanos, 2014, Bond supply and excess bond returns, *Review of Financial Studies* 27, 663–713.
- Hanson, S., 2014, Mortgage convexity, *Journal of Financial Economics* 113, 270–299.
- He, Z., B. Kelly, and A. Manela, 2017, Intermediary asset pricing: New evidence from many asset classes, *Journal of Financial Economics* 126, 1–35.
- Joslin, S., and Y. Konchitchki, 2018, Interest rate volatility, the yield curve, and the macroeconomy, *Journal of Financial Economics* 128, 344–362.
- Klingler, S., and S. Sundaresan, 2019, An explanation of negative swap spreads: demand for duration from underfunded pension plans, *Journal of Finance* 74, 675–710.
- Kondor, P., 2009, Risk in dynamic arbitrage: The price effects of convergence trading, *Journal of Finance* 64, 631–655.
- Mitchell, M., and T. Pulvino, 2012, Arbitrage crashes and the speed of capital, *Journal of Financial Economics* 104, 469–490.

- Realdon, M., 2024, The efficiency of the Estr overnight index swap market, *Journal of International Financial Markets, Institutions and Money* 91, 101943.
- Rebonato, R., and R. Ronzani, 2021, Is convexity efficiently priced? Evidence from international swap markets, *Journal of Empirical Finance* 63, 392–413.
- Rebonato, R., and V. Putyatin, 2018, The value of convexity: A theoretical and empirical investigation, *Quantitative Finance* 18, 11–30.
- Shleifer, A., and R. Vishny, 1997, The limits of arbitrage, *Journal of Finance* 52, 35–55.